

ORIGINAL ARTICLE

Technology-assisted risk of bias assessment in systematic reviews: a prospective cross-sectional evaluation of the RobotReviewer machine learning tool

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Abstract**Objectives:** To evaluate the reliability of RobotReviewer's risk of bias judgments.**Study Design and Setting:** In this prospective cross-sectional evaluation, we used RobotReviewer to assess risk of bias among 1,180 trials. We computed reliability with human reviewers using Cohen's kappa coefficient and calculated sensitivity and specificity. We investigated differences in reliability by risk of bias domain, topic, and outcome type using the chi-square test in meta-analysis.**Results:** Reliability (95% CI) was moderate for random sequence generation (0.48 [0.43, 0.53]), allocation concealment (0.45 [0.40, 0.51]), and blinding of participants and personnel (0.42 [0.36, 0.47]); fair for overall risk of bias (0.34 [0.25, 0.44]); and slight for blinding of outcome assessors (0.10 [0.06, 0.14]), incomplete outcome data (0.14 [0.08, 0.19]), and selective reporting (0.02 [−0.02, 0.05]). Reliability for blinding of participants and personnel ($P < 0.001$), blinding of outcome assessors ($P = 0.005$), selective reporting ($P < 0.001$), and overall risk of bias ($P < 0.001$) differed by topic. Sensitivity and specificity (95% CI) ranged from 0.20 (0.18, 0.23) to 0.76 (0.72, 0.80) and from 0.61 (0.56, 0.65) to 0.95 (0.93, 0.96), respectively.**Conclusion:** Risk of bias appraisal is subjective. Compared with reliability between author groups, RobotReviewer's reliability with human reviewers was similar for most domains and better for allocation concealment, blinding of participants and personnel, and overall risk of bias. © 2018 Elsevier Inc. All rights reserved.**Keywords:** Machine learning; Automation; Bias; Clinical trials; Evidence-based medicine; Evaluation

1. Introduction

Systematic reviews of well-designed randomized controlled trials provide the most robust evidence for assessing the effectiveness of health-care interventions [1–3] and are a foundation for evidence-based decision-making by numerous stakeholders. With the continued growth of evidence from clinical trials [4], the need for systematic reviews that transform this wealth of knowledge into a usable form is becoming increasingly pressing [3]. Because systematic

reviews require substantial intellectual manpower [2,5] and generally take at least 6 months to a year to produce [2,6], evidence is now accumulating more rapidly than it can be synthesized [4]. Systematic reviews thus quickly become out of date [5,7], and the inability to update them quickly enough contributes to research waste [5].

The living systematic review is an emerging method that aims to narrow the evidence-practice gap by increasing efficiencies in the systematic review process [3]. In contrast to the heavy workload required to update a traditional systematic review, living systematic reviews are housed online and updated in real time as new evidence emerges [3]. A living systematic review thus requires a more moderate and ongoing effort while aiming to conserve the rigor associated with traditional systematic reviews [3]. Maintaining living systematic reviews will require more rapid processes, and even within the context of traditional systematic reviews, strategies to modernize and accelerate the tasks associated with identifying and synthesizing evidence are

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What is new?**Key findings**

- Compared with reliability between author groups, RobotReviewer's reliability with human reviewers was comparable for most domains, and better for some domains.
- RobotReviewer's practical application may be improved by incorporating the ability to appraise all risk of bias domains as suggested by the Cochrane Collaboration.

What this adds to what was known?

- One previous study has evaluated RobotReviewer's accuracy compared with human reviewers.
- This study is the first to evaluate RobotReviewer's reliability, while accounting for agreement by chance.

What is the implication and what should change now?

- There is a need for guidance on how and when automated technologies may be applied to expedite the process of completing and updating knowledge syntheses.
- Continued evaluation of this and other machine learning technologies will add to the limited empirical evidence of their reliability and inform their practical application.

required [3]. Within the present research landscape, automation has the potential to play a key role in keeping systematic reviews up-to-date [8].

Certain essential yet time- and resource-intensive components of systematic reviews are particularly amenable to automation [8,9]. For example, judging the internal validity of the included trials requires the assessment of risk of bias among six domains and overall for each record [10]. Standard practice requires two reviewers to assess risk of bias and reach consensus [10] which can take 10 to 60 minutes per trial [11], representing a significant delay in the knowledge synthesis pipeline [8]. Viewing the assessment of risk of bias as a classification task and capitalizing on machine learning technologies, two tools have been developed [6,12] that reduce the time to assess risk of bias to only seconds. At the time of this study, one of these tools, RobotReviewer [12], assessed six of the risk of bias domains of the Cochrane Risk of Bias tool [10].

Before the use of automated technologies can be recommended in practice, evaluation of their ability to satisfactorily replicate the work of human reviewers is required. We

used trials covering 10 topics from systematic reviews or methodological research projects completed by our team to: (a) evaluate the accuracy (using percent agreement) and reliability (using Cohen's kappa coefficient) of RobotReviewer's risk of bias assessments compared with those of human reviewers, for each risk of bias domain and overall risk of bias, by systematic review topic and overall, (b) evaluate RobotReviewer's sensitivity and specificity for each risk of bias domain and overall risk of bias, by systematic review topic and overall, and (c) evaluate differences in accuracy and reliability across risk of bias domains and systematic review topics, and by the nature of the outcome assessed (objective or subjective).

2. Materials and methods**2.1. RobotReviewer**

RobotReviewer (<https://robot-reviewer.vortextext.systems/>) is a freely available, online machine learning tool developed by an international team of researchers to improve efficiency in the systematic review process [12]. It was trained to assess risk of bias using distant supervision and a heterogeneous sample of clinical trials ($n = 12,808$) from the Cochrane Database of Systematic Reviews [12]. As opposed to supervised machine learning which requires time-consuming and costly direct human supervision [12], distant supervision takes advantage of previously existing databases [13,14], facilitating a larger training data set and better performance of the model [15].

To assess risk of bias in a given trial, the user must drag and drop the document to the appropriate area on RobotReviewer's user interface. Risk of bias judgments of 'low' or 'high/unclear' are produced for the six core domains of the Cochrane Risk of Bias tool [10]: random sequence generation; allocation concealment; blinding of participants and personnel; blinding of outcome assessment; incomplete outcome data; and selective reporting. The system only assesses risk of bias from publications of trials published in English that are available in Portable Document Format [12]. The system also provides support for each judgment by extracting relevant text from the document and organizing it into a table akin to the 'risk of bias table' required in the reporting of Cochrane systematic reviews [10]. In real-life applications of RobotReviewer, the tool is designed to be used in a semiautomated fashion. By making use of the text extracted from each study, the human reviewer would have the opportunity to confirm or refute RobotReviewer's decisions. Importantly, in the present evaluation, we measured the reliability of the risk of bias judgments only, and not the reliability of the extracted text.

2.2. Included systematic reviews and research projects

We selected systematic reviews or methodological research projects covering 10 topics that had been

completed by our team at the Alberta Research Centre for Health Evidence and UAlberta Evidence-based Practice Centre, University of Alberta, Canada. Our sample included varied populations (i.e., adults and children) and intervention modalities (e.g., pharmacological, behavioral). The selected topics included: treatment of acute migraine (for children and adolescents [16], and adults [17]); antipsychotics (for children and young adults [18], and adults [19]); pharmacological treatments for bronchiolitis (for infants and young children) [20,21]; behavioral interventions for diabetes (for adults) [22,23]; rehydration treatments for gastroenteritis (for children) [24]; pain management for hip fracture (for adults) [25]; design, conduct, and reporting of child health trials [26]; quantifying bias in child health trials [27]; and validity and inter-rater reliability of quality assessment instruments [28].

All of the systematic reviews or methodological research projects were completed between 2011 and 2016. While undertaking the projects, we used the Cochrane Risk of Bias Tool [10] to assess risk of bias for all included trials among the six aforementioned core domains as well as other sources of bias (e.g., baseline imbalances in participant characteristics; study design characteristics). Overall risk of bias was deemed low if the trial was ranked low on all domains; it was deemed unclear if any domain was ranked unclear and none were ranked high; and it was deemed high if any domain was ranked high [10]. For all trials, risk of bias was assessed independently by two reviewers, who convened to discuss the assessments until consensus was reached or a third party provided arbitration. All reviewers were trained in the assessment of risk of bias with the support of internal guidance documents at our research center.

2.3. Data collection and analysis

2.3.1. Data collection

Between June and October 2016, we uploaded all the trials to RobotReviewer and recorded the automated risk of bias assessments in a Microsoft Office Excel (v. 2016; Microsoft Corporation, Redmond, WA) workbook. Only publications available in English and in Portable Document Format were included, though we did not exclude records by publication type. We compared RobotReviewer's assessments to the consensus of the two human reviewers and recorded the accuracy (i.e., if the assessment was the same or different) for each trial. Because RobotReviewer assesses risk of bias dichotomously, the human assessments were converted to dichotomous categories (i.e., 'high' or 'unclear' assessments were collapsed to 'high/unclear') for ease of comparison. Though we did not compare the assessment for other sources of bias as RobotReviewer does not have the capability to assess for these, the other bias domain was still considered in the judgment of overall risk of bias by the human reviewers. For a small subset of trials ($n = 215$, 18.2%) where human assessments were available

for both subjective and objective outcomes, we compared both of these to RobotReviewer's assessments (blinding and incomplete outcome data domains only).

Of note, the Cochrane Risk of Bias Tool underwent revision in 2010 [10]. Among other changes, blinding, which was previously assessed as a single domain, was separated into blinding of the participants and personnel and blinding of the outcome assessors. To account for this change, when blinding was assessed as a single domain by the human reviewers ($n = 435$ trials, 37%), the same judgment was used for both blinding of the participants and personnel and blinding of the outcome assessors for comparison to RobotReviewer.

2.3.2. Data analysis

The data were transferred to SPSS (v. 24; International Business Machines Corporation, Armonk, NY), Review Manager (v. 5.3; The Nordic Cochrane Centre, Copenhagen, Denmark), and StatXact (v. 10.0; Cytel Software Corporation, Cambridge, MA) for analysis. We used descriptive statistics (counts and percentages) to describe the sample of trials and risk of bias assessments, both overall and stratified by review topic and risk of bias domain. Differences in the descriptive characteristics of the included topics were assessed using the Fischer's exact test.

RobotReviewer's accuracy compared with the consensus of two human reviewers was tabulated using counts and percentages, for each risk of bias domain and overall risk of bias within each review topic and overall. To account for chance agreement, we tabulated reliability using Cohen's kappa coefficient and followed Landis and Koch's recommendations for interpretation (i.e., poor [$k < 0$]; slight [$k = 0.0-0.20$]; fair [$k = 0.21-0.40$]; moderate [$k = 0.41-0.60$]; substantial [$k = 0.61-0.80$]; or almost perfect [$k = 0.81-1.0$]) [29]. For trials where risk of bias assessments were completed for both objective and subjective outcomes, only the results for the objective outcomes were included in the overall findings to avoid duplicates in the random sequence generation and allocation concealment domains. In our analyses, we found no statistical difference in the reliability between RobotReviewer's risk of bias appraisals and those of the human reviewers for objective versus subjective outcomes; using the appraisals for the objective or subjective outcomes in the overall findings would therefore yield extremely similar results. Statistical differences in accuracy and reliability across systematic review topics (by domain), between risk of bias domains (overall and within each topic), and by outcome type (objective or subjective) were assessed using the chi-square test in meta-analysis.

We calculated RobotReviewer's sensitivity and specificity to determine the tool's ability to correctly identify domains at low or at unclear or high risk of bias. We used values from cross-tabulations to calculate (a) sensitivity and (b) specificity using the following respective formulas:

- (a) Sensitivity = # true positives/(# true positives + # false negatives), and
 (b) Specificity = # true negatives/(# true negatives + # false positives)

True positives and negatives occurred when RobotReviewer correctly judged a domain as low or high/unclear risk, respectively (i.e., in agreement with the human reviewers). False positives and negatives occurred when RobotReviewer incorrectly judged a domain as low or high/unclear risk, respectively (i.e., in disagreement with the human reviewers). Sensitivity and specificity for each risk of bias domain were depicted visually using receiver operating characteristics plots.

3. Results

3.1. Sample characteristics

Of the 1,323 trials included in the sample, a total of 1,180 were in a format that could be recognized by RobotReviewer (Table 1). Detailed characteristics of the sample are in Appendix A. Of the records that RobotReviewer could not assess ($n = 143$; 10.8%), 13 (9.1%) were not published in English and 130 (90.9%) were not in a format that the machine could recognize (e.g., scanned images of records not accessible online). Across topics, a significantly greater proportion of the trials for bronchiolitis (9.1%) and for hip fracture (7.6%) were not English as compared with the others ($P < 0.001$). A significantly greater proportion of trials in quantifying bias in trials (33.8%) could not be assessed because they were not in a format that RobotReviewer could recognize ($P < 0.001$). These were mainly older articles that had been scanned from physical copies of academic journals. The vast majority of records were published research articles ($n = 1,159$; 98.2%), though a small number of abstracts ($n = 5$; 0.4%), protocols ($n = 2$; 0.2%), registries ($n = 7$; 0.6%), reports ($n = 4$; 0.3%), and theses ($n = 3$; 0.3%) were also included.

3.2. Accuracy

A descriptive summary of the accuracy of RobotReviewer's risk of bias assessments as compared with those of the human reviewers is shown in Table 2. A detailed summary of RobotReviewer's accuracy by topic is in Appendix B. Accuracy differed significantly across domains, both overall ($P < 0.001$) and within topics ($P < 0.05$) with the exception of bronchiolitis ($P = 0.970$) where agreement was $\geq 69\%$ for all domains. Combining the data from all topics, accuracy was best for overall risk of bias (90.6%; range 67.4–100.0%), followed by allocation concealment (77.3%; 68.4–95.7%), blinding of participants and personnel (75.5%; 41.4–82.4%), random sequence generation (73.5%; 69.0–85.3%), incomplete outcome data (53.7%; 33.7–75.0%), blinding of

outcome assessors (43.5%; 2.6–75.0%), and selective reporting (38.6%; 6.0–74.4%). Accuracy differed significantly across topics for allocation concealment ($P < 0.001$), blinding of participants and personnel ($P < 0.001$), blinding of the outcome assessors ($P < 0.001$), incomplete outcome data ($P < 0.001$), selective reporting ($P < 0.001$), and overall risk of bias ($P < 0.001$), but not for random sequence generation ($P = 0.16$).

3.3. Reliability

Reliability for each domain is shown in Table 3. Detailed reliability data for each topic are in Appendix C. Reliability differed significantly across domains, both overall ($P < 0.001$) and within topics ($P < 0.05$) with the exception of bronchiolitis ($P = 0.080$) and gastroenteritis ($P = 0.740$). Combining the data from all topics, reliability (k [95% CI]) was moderate for random sequence generation (0.48 [0.43, 0.53]), allocation concealment (0.45 [0.40, 0.51]), and blinding of participants and personnel (0.42 [0.36, 0.47]). Reliability was fair for overall risk of bias (0.34 [0.25, 0.44]), and slight for incomplete outcome data (0.14 [0.08, 0.19]), blinding of outcome assessors (0.10 [0.06, 0.14]), and selective reporting (0.02 [−0.02, 0.05]). Reliability differed significantly across topics for blinding of participants and personnel ($P < 0.001$), blinding of outcome assessors ($P = 0.005$), selective reporting ($P < 0.001$), and overall risk of bias ($P < 0.001$). There were no differences across topics with respect to reliability for random sequence generation ($P = 0.250$), allocation concealment ($P = 0.550$), and incomplete outcome data ($P = 0.170$). In the comparison of reliability for the two systematic reviews with both objective and subjective outcomes, no differences by outcome were observed ($P > 0.05$; Table 4). Detailed reliability data by review topic are in Appendix D.

3.4. Sensitivity and specificity

RobotReviewer's sensitivity (i.e., true positive rate, or the rate at which RobotReviewer correctly identified domains at low risk of bias) and specificity (i.e., true negative rate, or the rate at which RobotReviewer correctly identified domains at high/unclear risk of bias) are shown in Table 3. Detailed sensitivity and specificity data for each topic are in Appendix C. Combining the data from all topics, sensitivity (95% CI) was best for random sequence generation (0.76 [0.72, 0.80]), followed by allocation concealment (0.64 [0.58, 0.69]), incomplete outcome data (0.53 [0.49, 0.57]), blinding of participants and personnel (0.47 [0.42, 0.52]), overall risk of bias (0.34 [0.23, 0.46]), blinding of outcome assessors (0.28 [0.25, 0.32]), and selective reporting (0.20 [0.18, 0.23]). Specificity (95% CI) was best for overall risk of bias (0.95 [0.93, 0.96]), followed by blinding of participants and personnel (0.90

Table 1. Characteristics of the included systematic reviews and methodological research projects

Topic	Included records (<i>N</i>)	Included document type, <i>n</i> (%)	
		Full-text articles	Other ^a
Acute migraine	95	86 (90.5)	9 (9.5)
Antipsychotics in adults	116	116 (100.0)	0 (0.0)
Antipsychotics in children	58	53 (91.4)	5 (8.6)
Behavioral diabetes	157	154 (98.1)	3 (1.9)
Bronchiolitis	39	39 (100.0)	0 (0.0)
Design, conduct, and reporting of trials	290	290 (100.0)	0 (0.0)
Gastroenteritis	28	28 (100.0)	0 (0.0)
Hip fracture	57	53 (93.0)	4 (7.0)
Quantifying bias in trials	187	187 (100.0)	0 (0.0)
Validity and inter-rater reliability	153	153 (100.0)	0 (0.0)
Overall	1,180	1,159 (98.2)	21 (1.8)

^a Abstracts, protocols, registries, reports, and theses.

[0.87, 0.92]), allocation concealment (0.82 [0.79, 0.85]), blinding of outcome assessors (0.82 [0.78, 0.85]), selective reporting (0.81 [0.77, 0.85]); random sequence generation (0.72 [0.68, 0.75]), and incomplete outcome data (0.61 [0.56, 0.65]). Receiver operating characteristics plots illustrating sensitivity and specificity for each topic with each risk of bias domain are shown in [Appendix E](#).

4. Discussion

Given the volume of accumulating scientific evidence in the form of randomized controlled trials, the need for systematic reviews to synthesize this evidence into usable information for clinicians is clear [4]. Because many of the processes involved in knowledge synthesis are demanding on time and resources (e.g., record screening, data extraction, quality assessment), systematic reviews are often out of date before they are even published [5,7] and rarely updated when new evidence emerges [5]. To date, more than 100 automated technologies have been developed to expedite some of the most time-consuming steps in the knowledge-synthesis process [30]. In the present study, we tested the reliability of one tool, RobotReviewer, which aims to expedite the assessment of risk of bias in randomized controlled trials. RobotReviewer minimized the time to complete risk of bias assessments to seconds or less per trial. Reliability compared with human reviewers was slight to moderate depending on the domain assessed, though performance differed significantly by topic within

most domains. RobotReviewer was more specific than it was sensitive (i.e., was better able to correctly identify trials at unclear or high than at low risk of bias) for allocation concealment, blinding of participants and personnel and of the outcome assessors, selective reporting, and overall risk of bias. It was nearly as sensitive as it was specific for random sequence generation and incomplete outcome data. The tool is still in development, and further rigorous evaluations will be necessary as it continues to evolve to determine its practical utility in real-world knowledge-synthesis projects. Should this and other automated technologies prove reliable and practical, substantial time and resource savings could be realized and research waste could be reduced [5].

The assessment of risk of bias is inherently subjective, and even between individual human reviewers and independent author groups, the Cochrane Risk of Bias Tool has been criticized for its poor reliability [31,32]. Marshall et al. [12] previously evaluated RobotReviewer and reported accuracy compared with the consensus of human reviewers, compared with the agreement between independent author groups (*n* = 6,610 trials). Agreement between author groups ranged from 71 to 85% (random sequence generation, 85%; allocation concealment, 80%; blinding of participants and personnel, 78%; blinding of outcome assessors, 83%; incomplete outcome data, 71%; and selective reporting, 73%) [12]. Despite exceeding 60% for all domains (random sequence generation, 74%;

Table 2. Accuracy of RobotReviewer's risk of bias assessments compared with the consensus of two human reviewers for *N* = 1,180 trials

Risk of bias domain	Assessed as low risk of bias, <i>n</i> (%)		Accurate assessments, <i>n</i> (%) ^a
	Human reviewers	RobotReviewer	
Random sequence generation	496 (42.0)	573 (48.6)	867 (73.5)
Allocation concealment	321 (27.2)	367 (31.1)	912 (77.3)
Blinding: participants and personnel	406 (34.4)	291 (24.7)	891 (75.5)
Blinding: outcome assessors	693 (58.7)	302 (25.6)	513 (43.5)
Incomplete outcome data	674 (57.1)	565 (47.9)	634 (53.7)
Selective reporting	834 (70.7)	255 (21.6)	456 (38.6)
Overall risk of bias	82 (6.9)	101 (8.6)	1,069 (90.6)

^a Accurate assessments were those where RobotReviewer judged the risk of bias to be the same as the human reviewers.

Table 3. RobotReviewer's reliability, sensitivity, and specificity compared with risk of bias assessments completed by human reviewers for $N = 1,180$ trials

Risk of bias domain	Cohen's kappa (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
Random sequence generation	0.48 (0.43, 0.53)	0.76 (0.72, 0.80)	0.72 (0.68, 0.75)
Allocation concealment	0.45 (0.40, 0.51)	0.64 (0.58, 0.69)	0.82 (0.79, 0.85)
Blinding: participants and personnel	0.42 (0.36, 0.47)	0.47 (0.42, 0.52)	0.90 (0.87, 0.92)
Blinding: outcome assessors	0.10 (0.06, 0.14)	0.28 (0.25, 0.32)	0.82 (0.78, 0.85)
Incomplete outcome data	0.14 (0.08, 0.19)	0.53 (0.49, 0.57)	0.61 (0.56, 0.65)
Selective reporting	0.02 (−0.02, 0.05)	0.20 (0.18, 0.23)	0.81 (0.77, 0.85)
Overall risk of bias	0.34 (0.25, 0.44)	0.34 (0.23, 0.46)	0.95 (0.93, 0.96)

Abbreviation: CI = confidence interval.

allocation concealment, 74%; blinding of participants and personnel, 73%; blinding of outcome assessors, 62%; incomplete outcome data, 65%; selective reporting, 68%), RobotReviewer's accuracy compared with human consensus was significantly less than agreement between author groups [12]. Though not tested statistically, in our sample, RobotReviewer's accuracy was similar to that reported by Marshall et al. [12] for random sequence generation (74%), allocation concealment (77%), and blinding of participants and personnel (76%). Compared to the findings of Marshall et al. [12], we found 18%, 11%, and 29% lesser accuracy for blinding of the outcome assessors (44%), incomplete outcome data (54%), and selective reporting (39%), respectively.

During the tool's development, Marshall et al. [12] investigated a baseline model, where all test documents were labeled with the majority class (high/unclear or low) for each domain. Accuracy for the model was 59% for random sequence generation, 54% for allocation concealment, 50% for blinding of participants and personnel, 58% for blinding of the outcome assessors, 61% of incomplete outcome reporting, and 60% for selective reporting [12]. The accuracy of the tool improved as more information was incorporated into the model (e.g., sentence relevance). Interestingly, we calculated better accuracy compared to the baseline model for random sequence generation (74%), allocation concealment (77%), and blinding of participants and personnel (76%). For the blinding of the outcome assessors (44%), incomplete outcome data (54%), and selective reporting (39%) domains, accuracy for our sample was in fact inferior to that reported for the baseline model of Marshall et al. [12].

Accuracy is strongly influenced by the distribution of risk of bias judgments, so accounting for chance agreement

was important when evaluating RobotReviewer's performance. In our study, we accounted for chance agreement by calculating reliability using Cohen's kappa statistic. Studies that have investigated the reliability of the Risk of Bias Tool between independent author groups [31,32] have found it to be moderate to substantial for random sequence generation, fair for allocation concealment, slight for blinding of participants and personnel, selective reporting, and overall risk of bias, and slight to fair for incomplete outcome data. Using a heterogeneous sample of randomized controlled trials, we found RobotReviewer's reliability to be comparable to that between independent author groups [31,32] for random sequence generation, incomplete outcome data, and selective reporting, for which agreement with human consensus was slight to moderate. RobotReviewer's fair-to-moderate reliability with human consensus for allocation concealment, blinding of participants and personnel, and overall risk of bias exceeded previously reported levels of reliability between independent author groups within these domains [31,32]. RobotReviewer's performance was particularly impressive in light of the fact that we evaluated its ability to fully automate the assessment of risk of bias, whereas it was designed to be a semiautomated tool (see 2.1).

Despite comparable performance with assessments completed by independent author groups, RobotReviewer presents a number of limitations that will need to be overcome to facilitate its use in the real world. First, the tool only assesses risk of bias as 'low' or 'high/unclear', which is not consistent with the guidance laid out in the Cochrane Handbook [10]. Moreover, at the time of data collection, RobotReviewer did not have the capability to assess other sources of bias, a necessary contributor to the judgment of overall risk of bias [10]. When the

Table 4. Reliability, sensitivity, and specificity for $n = 215$ trials where risk of bias was assessed for both subjective and objective outcomes

Metric	Blinding: participants and personnel		Blinding: outcome assessors		Incomplete outcome data	
	Objective	Subjective	Objective	Subjective	Objective	Subjective
Cohen's kappa (95% CI)	0.18 (0.03, 0.33)	0.27 (0.08, 0.46)	0.03 (0.00, 0.06)	−0.08 (−0.21, 0.05)	0.06 (−0.09, 0.22)	0.07 (−0.09, 0.22)
Sensitivity (95% CI)	0.17 (0.07, 0.32)	0.23 (0.10, 0.42)	0.17 (0.11, 0.24)	0.10 (0.02, 0.27)	0.32 (0.20, 0.45)	0.32 (0.20, 0.47)
Specificity (95% CI)	0.97 (0.92, 0.99)	0.97 (0.92, 0.99)	0.94 (0.70, 1.00)	0.83 (0.75, 0.89)	0.75 (0.65, 0.82)	0.74 (0.65, 0.82)

Abbreviation: CI = confidence interval.

website was revisited in April 2017, the tool was only producing assessments for four domains (random sequence generation, allocation concealment, blinding of participants and personnel, and blinding of outcome assessors). Realistically some domains may prove less amenable to automation. For example, judging attrition bias requires the calculation of the proportion of participants lost to follow-up, whereas judging reporting bias requires the consultation of additional information sources beyond the published trial (e.g., protocols, trial registries, contacting the corresponding author) [10]. RobotReviewer does not yet support the incorporation of additional calculations or information sources into its risk of bias judgments. For these reasons, since the time of our investigation appraisals for the selective reporting and incomplete outcome data domains have been removed from the RobotReviewer tool [12].

At the present time, RobotReviewer provides risk of bias judgments per unit trial and does not have the capability to assess risk of bias for more than one outcome per trial. Cochrane guidelines suggest that individual judgments may be made for each outcome, or group of outcomes (i.e., objective and subjective), within each trial for the blinding and incomplete outcome data domains [10]. Still, even when multiple outcomes are investigated within a systematic review, the majority of review authors assess risk of bias at the study level, rather than the outcome level [33]. The same is likely true of the reviews in the data set that was used to train RobotReviewer. Our analysis of a small subsample of trials did not show a statistical difference in reliability when human assessments for both objective and subjective outcomes were compared with RobotReviewer's singular risk of bias judgments. Nevertheless, evidence indicates that in trials of clinical interventions, bias in the magnitude of the treatment effect (e.g., from inadequate blinding or allocation concealment) varies depending on the type of outcome measured [34].

Another serious limitation of the tool was its inability to assess trials that had not been published online (i.e., when only scanned images of the physical articles were available). This was commonly the case in systematic reviews where older articles were included, and in some systematic reviews or research projects this may severely reduce the tool's practical utility. For example, in our study of bias in trials [27], more than one-third of the records could not be assessed by RobotReviewer. Moreover, at the present time RobotReviewer only has the capability to assess articles published in English. A recent investigation of child health systematic reviews found that most reviewers searched for non-English studies [35]. Empirical evidence indicates that excluding non-English records has the potential to impact the magnitude of effect in systematic reviews of complementary and alternative medicine [36] and child health [35] interventions.

4.1. Implications and future directions

Despite the proliferation of technologies to support and expedite the most arduous of systematic review tasks [30], to date their reliability and usability have not been extensively studied. As such, their practical significance remains uncertain. The completion of high-quality systematic reviews requires adherence to rigorous published conduct and reporting guidelines [10,37], within which there is currently no indication for the use of automation. Nevertheless, high-standard practices may be impractical in fields where new evidence from published trials is accumulating quickly, or when evidence to support decision-making is urgently required. Presently and in the future, knowledge synthesis products will need to be produced more rapidly, and streamlined processes will be necessary to keep them up-to-date. Overall quality will need to be balanced with efficiency and practicality to keep pace with the rate at which research knowledge is accumulating [4]. Guidance for the rigorous completion and reporting of rapid review products and living systematic reviews will need to be further developed and should include advice for reviewers with respect to when and how automated technologies could be applied.

With respect to RobotReviewer, the tool is relatively new (documentation of its development and early evaluation was first published in 2016 [12]) and still in development. Building on these and the findings of future evaluations, it is likely that its capabilities will continue to grow and its reliability will continue to improve. In fact, since the time of this evaluation, RobotReviewer's underlying machine learning algorithms have been updated to use a neural network-based approach [38,39], which has provided some improvements in performance. In future revisions to the tool, provisions to allow for outcome-level assessments of risk of bias should be strongly considered. Also, maximizing the tool's capability to assess risk of bias in the array of trials potentially included in a systematic review, including older trials and those not published in English, would improve its practical appeal. The continued evaluation of the reliability and usability of tools like RobotReviewer among large, heterogeneous samples of trials will provide empirical data to inform their suitability and application in practice.

4.2. Strengths and limitations

To our knowledge, this is only the second study to investigate the accuracy and reliability of RobotReviewer's risk of bias assessments compared with the consensus of human reviewers, and the first by authors outside of the tool development team. As there is no standard for an acceptable level of reliability for risk of bias judgments between reviewers, we compared RobotReviewer's judgments to the consensus of human reviewers, which is imperfect. Nevertheless, the consensus of two human reviewers is the best available standard for risk of bias, which is inherently subjective.

We used a relatively large, heterogeneous sample of trials, including varied intervention modalities and populations. Of note, we included three Cochrane systematic reviews in our sample. We cannot ascertain whether the trials included in these systematic reviews were also included within the sample of trials used to train RobotReviewer, or the sample of trials used by the developers to test its accuracy [12]. The same is true for the other trials included in our sample, which had the potential to be the same as those used to develop RobotReviewer (although the risk of bias appraisals would have been our team's own, so could have differed from those in Cochrane systematic reviews). Moreover, we included a range of document types in our sample, whereas RobotReviewer was trained using published trial reports only. In real-life applications, authors will need to appraise trials that may have been used for RobotReviewer's training, as well as those available as different document types (abstracts, dissertations, trial registers). Our evaluation therefore provides a realistic appraisal of the tool's reliability.

We noted significant differences in accuracy and reliability across topics, so the overall findings may not be generalized to all topics. Further investigation is required to identify topic-related variables that may contribute to differences in performance. For the comparison of trials in which risk of bias was assessed for both objective and subjective outcomes, we used a relatively small sample from only two topics. Findings for the two blinding domains need to be interpreted with prudence, keeping in mind the revisions to the Cochrane Risk of Bias Tool in 2010 which could have impacted the outcomes.

5. Conclusions

Automated technologies have been developed with the aim of expediting steps in the systematic review process. Our evaluation of RobotReviewer revealed that, within a heterogeneous sample of trials, its reliability was at least comparable to that between independent author groups for all risk of bias domains. Nevertheless, the tool's implementation in practice is hindered by design limitations that preclude its ability to assess overall risk of bias: some core domains are not available and judgments are provided only dichotomously. As this and other automated technologies are novel and still developing, ongoing evaluation of their reliability and practicability will be important. Such evaluations will provide information for their continued improvement, and the empirical evidence required to inform guidance for their use.

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2017.12.015>.

References

- [1] McNair P, Lewis G. Levels of evidence in medicine. *Int J Sports Phys Ther* 2012;7(5):474–81.
- [2] Askie L, Offringa M. Systematic reviews and meta-analysis. *Semin Fetal Neonatal Med* 2015;20(6):403–9.
- [3] Elliott JH, Turner T, Clavisi O, Thomas J, Higgins JP, Mavergames C, et al. Living systematic reviews: an emerging opportunity to narrow the evidence-practice gap. *PLoS Med* 2014;11(2):e1001603.
- [4] Bastian H, Glasziou P, Chalmers I. Seventy-five trials and eleven systematic reviews a day: how will we ever keep up? *PLoS Med* 2010;7(9):e1000326.
- [5] Crequit P, Trinquart L, Yavchitz A, Ravaud P. Wasted research when systematic reviews fail to provide a complete and up-to-date evidence synthesis: the example of lung cancer. *BMC Med* 2016;14:8.
- [6] Millard LA, Flach PA, Higgins JP. Machine learning to assist risk-of-bias assessments in systematic reviews. *Int J Epidemiol* 2016;45:266–77.
- [7] Shojania KG, Sampson M, Ansari MT, Ji J, Doucette S, Moher D. How quickly do systematic reviews go out of date? A survival analysis. *Ann Intern Med* 2007;147:224–33.
- [8] Wallace BC, Dahabreh IJ, Schmid CH, Lau J, Trikalinos TA. Modernizing the systematic review process to inform comparative effectiveness: tools and methods. *J Comp Eff Res* 2013;2(3):273–82.
- [9] Tsertsvadze A, Chen YF, Moher D, Sutcliffe P, McCarthy N. How to conduct systematic reviews more expeditiously? *Syst Rev* 2015;4:160.
- [10] The Cochrane Collaboration, Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S, editors. *Cochrane handbook of systematic reviews*. London, UK: Cochrane; 2011. Version 5.1.0.
- [11] Savovic J, Weeks L, Sterne JA, Turner L, Altman DG, Moher D, et al. Evaluation of the Cochrane Collaboration's tool for assessing the risk of bias in randomized trials: focus groups, online survey, proposed recommendations and their implementation. *Syst Rev* 2014;3:37.
- [12] Marshall IJ, Kuiper J, Wallace BC. RobotReviewer: evaluation of a system for automatically assessing bias in clinical trials. *J Am Med Inform Assoc* 2016;23:193–201.
- [13] Mintz M, Bills S, Snow R, Jurafsky D. Distant supervision for relation extraction without labeled data. *Proceedings of the 47th meeting of the ACL and the 4th IJCNLP of the AFNLP*. Suntec, Singapore: Association for Computational Linguistics; 2009:1003–11.
- [14] Ling X, Jiang J, He X, Mei Q, Zhai C, Schatz B. Automatically generating gene summaries from biomedical literature. *Pac Symp Biocomput* 2006;40–51.
- [15] Banko M, Brill E. Scaling to very very large corpora for natural language disambiguation. *Proceedings of the 39th Annual Meeting on Association for Computational Linguistics*. Toulouse, France: Association for Computational Linguistics; 2001:26–33.
- [16] Richer L, Billingham L, Linsdell MA, Russell K, Vandermeer B, Crumley ET, et al. Drugs for the acute treatment of migraine in children and adolescents. *Cochrane Database Syst Rev* 2016;4:CD005220.
- [17] Sumamo Schellenberg E, Dryden DM, Pasichnyk D, Ha C, Vandermeer B, Friedman BW, et al. Acute migraine treatment in emergency settings. Rockville, MD: Agency for Healthcare Research and Quality; 2012.
- [18] Seida JC, Schouten JR, Mousavi SS, Hamm M, Beath A, Vandermeer B, et al. First- and second-generation antipsychotics for children and young adults. Rockville, MD: Agency for Healthcare Research and Quality; 2012.

- [19] Abou-Setta AM, Mousavi SS, Spooner C, Schouten JR, Pasichnyk D, Armijo-Olivo S, et al. First-generation versus second-generation antipsychotics in adults: comparative effectiveness. Rockville, MD: Agency for Healthcare Research and quality; 2012.
- [20] Hartling L, Bialy LM, Vandermeer B, Tjosvold L, Johnson DW, Plint AC, et al. Epinephrine for bronchiolitis. *Cochrane Database Syst Rev* 2011. CD003123.
- [21] Fernandes RM, Hartling L. Glucocorticoids for acute viral bronchiolitis in infants and young children. *JAMA* 2014;311:87–8.
- [22] Pillay J, Armstrong MJ, Butalia S, Donovan LE, Sigal RJ, Chordiya P, et al. Behavioral programs for type 1 diabetes mellitus: a systematic review and meta-analysis. *Ann Intern Med* 2015;163: 836–47.
- [23] Pillay J, Armstrong MJ, Butalia S, Donovan LE, Sigal RJ, Vandermeer B, et al. Behavioral programs for type 2 diabetes mellitus: a systematic review and network meta-analysis. *Ann Intern Med* 2015;163:848–60.
- [24] Freedman SB, Pasichnyk D, Black KJL, Fitzpatrick E, Gouin S, Milne A, et al. Pediatric Emergency Research Canada Gastroenteritis Study Group. Gastroenteritis therapies in developed countries: systematic review and meta-analysis. *PLoS One* 2015;12:e0176741.
- [25] Abou-Setta AM, Beaupre LA, Jones CA, Rashid S, Hamm MP, Sadowski CA, et al. Pain management interventions for hip fracture. Rockville, MD: Agency for Healthcare Research and Quality; 2011.
- [26] Gates A, Hartling L, Vandermeer B, Caldwell P, Contopoulos-Ioannidis DG, Curtis S, et al. The conduct and reporting of child health research: an analysis of randomized controlled trials published in 2012 and evaluation of change over 5 years. *J Pediatr* 2018;193: 237–44.e37.
- [27] Hartling L, Hamm MP, Fernandes RM, Dryden DM, Vandermeer B. Quantifying bias in randomized controlled trials in child health: a meta-epidemiological study. *PLoS One* 2014;9:e88008.
- [28] Hartling L, Hamm M, Milne A, Vandermeer B, Santaguida PL, Ansari M, et al. Validity and inter-rater reliability testing of quality assessment instruments. Rockville, MD: Agency for Healthcare Research and Quality; 2012.
- [29] Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics* 1977;33:159–74.
- [30] Marshall C. The systematic review toolbox. Available at <http://systematicreviewtools.com/index.php>. Accessed April 25, 2017.
- [31] Armijo-Olivo S, Ospina M, da Costa BR, Egger M, Saltaji H, Fuentes J, et al. Poor reliability between Cochrane reviewers and blinded external reviewers when applying the Cochrane risk of bias tool in physical therapy trials. *PLoS One* 2014;9:e96920.
- [32] Hartling L, Hamm MP, Milne A, Vandermeer B, Santaguida PL, Ansari M, et al. Testing the risk of bias tool showed low reliability between individual reviewers and across consensus assessments of reviewer pairs. *J Clin Epidemiol* 2013;66:973–81.
- [33] Katikireddi SV, Egan M, Petticrew M. How do systematic reviews incorporate risk of bias assessments into the synthesis of evidence? A methodological study. *J Epidemiol Community Health* 2015;69: 189–95.
- [34] Wood L, Egger M, Gluud LL, Schulz KF, Jüni P, Altman DG, et al. Empirical evidence of bias in treatment effect estimates in controlled trials with different interventions and outcomes: meta-epidemiological study. *BMJ* 2008;336:601–5.
- [35] Hartling L, Featherstone R, Nuspl M, Shave K, Dryden DM, Vandermeer B. Grey literature in systematic reviews: a cross-sectional study of the contribution of non-English reports, unpublished studies and dissertations to the results of meta-analyses in child-relevant reviews. *BMC Med Res Methodol* 2017;17:64.
- [36] Pham B, Klassen TP, Lawson ML, Moher D. Language of publication restrictions in systematic reviews gave different results depending on whether the intervention was conventional or complementary. *J Clin Epidemiol* 2005;58:769–76.
- [37] Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med* 2009;6(7):e1000097.
- [38] Zhang Y, Marshall I, Wallace BC. Rationale-augmented convolutional neural networks for text classification. *Proc Conf Empir Methods Nat Lang Process* 2016;2016:795–804.
- [39] Marshall IJ, Kuiper J, Banner E, Wallace BC. Automating biomedical evidence synthesis: RobotReviewer. *Proceeding of the 55th Annual Meeting of the Association for Computational Linguistics-System Demonstrations*. Vancouver, Canada: Association for Computational Linguistics; 2017:7–12.